

Your AI Career Begins Here

20 Prompt Engineering Interview Questions

You Can't Ignore.



1. What is prompt engineering?

Prompt engineering means writing good instructions for an AI model so it knows exactly what task to perform. When the prompt is clear, the model gives better answers. It's like giving proper directions so the AI does not get confused.

2. What is a zero-shot prompt? Give an example.

A zero-shot prompt is when you ask the AI to do something without giving any examples.

Example:

Prompt: "Identify the emotion of this sentence: 'I loved the food!' Emotion:"

Output: Positive

3. What does the temperature setting do in an LLM?

Temperature controls how "creative" or "random" the model's answer is.

- Low temperature (0–0.3) → predictable and safe answers
- High temperature (0.7–1.0) → more creative or varied answers

4. What is few-shot prompting and why do we use it?

Few-shot prompting means giving the model a few examples before asking it to do the task.

This helps the model understand the format or pattern you want.

We use few-shot prompting when:

- The task is a bit complex
- Zero-shot prompts don't give good results

5. What is the difference between system prompt and role prompt?

- System Prompt – Sets rules or instructions (e.g., "You must answer in bullet points").
- Role Prompt – Gives the AI a job or role (e.g., "Act like a friendly tutor").

Both help the model behave in a specific way.

6. What is top-K and top-P sampling?

These settings control how the AI chooses words.

- Top-K: Picks from the top K most likely words.
- Top-P: Picks from a group of words whose total probability adds up to P (like 0.9).

Both help control creativity and randomness.

7. What is Chain of Thought (CoT) prompting?

Chain of Thought prompting tells the AI to show its step-by-step thinking before giving the final answer.

This is useful for math, logic, puzzles, or anything needing reasoning.

8. What is self-consistency prompting?

Self-consistency means generating multiple reasoning paths and choosing the answer that appears most often.

It improves accuracy because it compares several possible solutions.

9. What is Tree of Thoughts (ToT)?

Tree of Thoughts is like Chain of Thought but instead of one line of thinking, the model explores many different branches.

This is helpful for planning, brainstorming, or solving creative problems.

10. What is ReAct prompting?

ReAct means Reason + Act.

The model thinks step-by-step and can also take actions—like searching the web, using tools, or performing calculations.

This technique is used to build AI agents

11. What is Automatic Prompt Engineering (APE)?

Automatic Prompt Engineering is when an AI helps create or improve prompts.

The model generates many versions of a prompt, tests them, and picks the one that works best.

This saves time and helps you discover prompts you may not think of manually.

12. What are some best practices for writing good prompts?

Good practices include:

- Write clear and simple instructions
- Mention the output format (bullet points, JSON, etc.)
- Give examples if needed
- Use the right tone (formal, friendly, etc.)
- Keep prompts short but specific

13. How can you design a prompt to extract structured data from messy text?

Tell the AI exactly what structure you want—for example, a JSON format or a list.

Also provide the fields clearly.

Example:

“Extract the product name and price. Return the result in JSON with keys: product and price.”

This makes the output more organized

14. What should you do if the model repeats words or generates loops?

To fix repetition or loops, try:

- Lowering temperature
- Increasing top-K or top-P slightly
- Telling the model to be brief
- Adding a “stop” word or stop sequence
- Splitting the task into smaller chunks

15. How do you handle long or multi-step tasks using prompts?

Break the task into smaller parts, such as:

1. Ask the AI to summarize
2. Then ask it to extract key points
3. Then ask it to generate final output

Working step-by-step makes complex tasks more reliable.

16. How can you fix hallucinations or inaccurate answers?

Try these approaches:

- Give clearer instructions
- Add examples
- Reduce temperature
- Provide reference text
- Use a structured output format
- Make the task smaller
- Double-check token limits (in API calls)

These reduce the chances of wrong or made-up information.

17. How can you use prompts to generate user interfaces or structured outputs like JSON or HTML?

You can tell the AI exactly which format you want.

Examples:

- "Create an HTML code for a login form."
- "Return the result as a JSON object with fields: title, date, and summary."

Giving a clear schema helps the model follow the structure perfectly.

18. What is the difference between Chain of Thought and ReAct prompting?

- Chain of Thought (CoT): The model writes out its step-by-step reasoning.
- ReAct: The model reasons and takes actions (like searching the web or using tools).

ReAct is more powerful when the task needs external information.

19. How does Tree of Thoughts (ToT) improve on Chain of Thought?

Tree of Thoughts explores many possible reasoning paths, not just one.

It evaluates each branch and picks the best direction.

This works well for puzzles, planning, and creative tasks.

20. What is multimodal prompting?

Multimodal prompting means using more than one type of input, like text + images, text + audio, or text + video.

Examples:

- "Describe what is happening in this image."
- "Extract text from this picture."
- It is used in tasks like captioning images, reading documents, or analyzing charts.

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Chandra

Senior GenAI Engineer